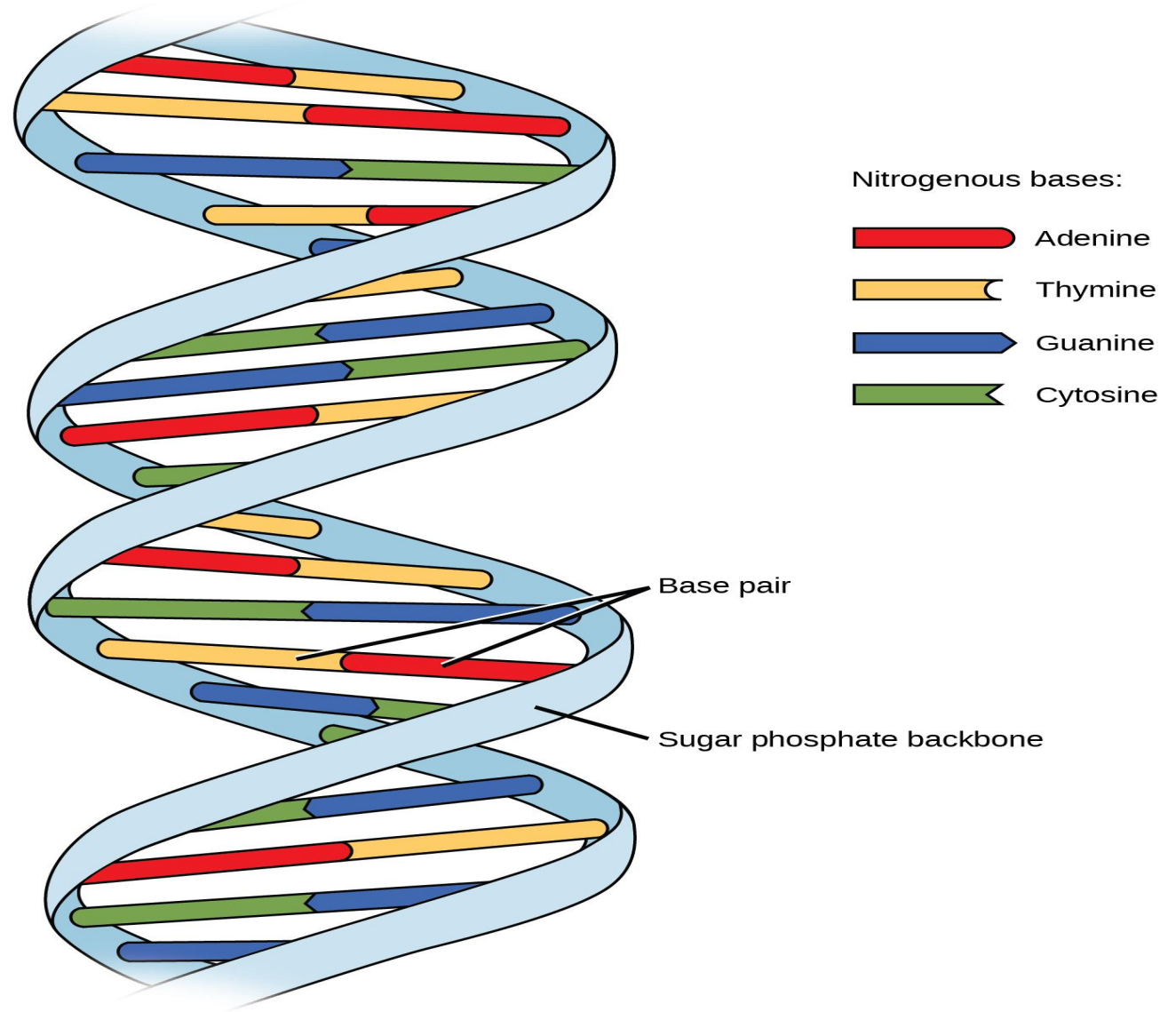


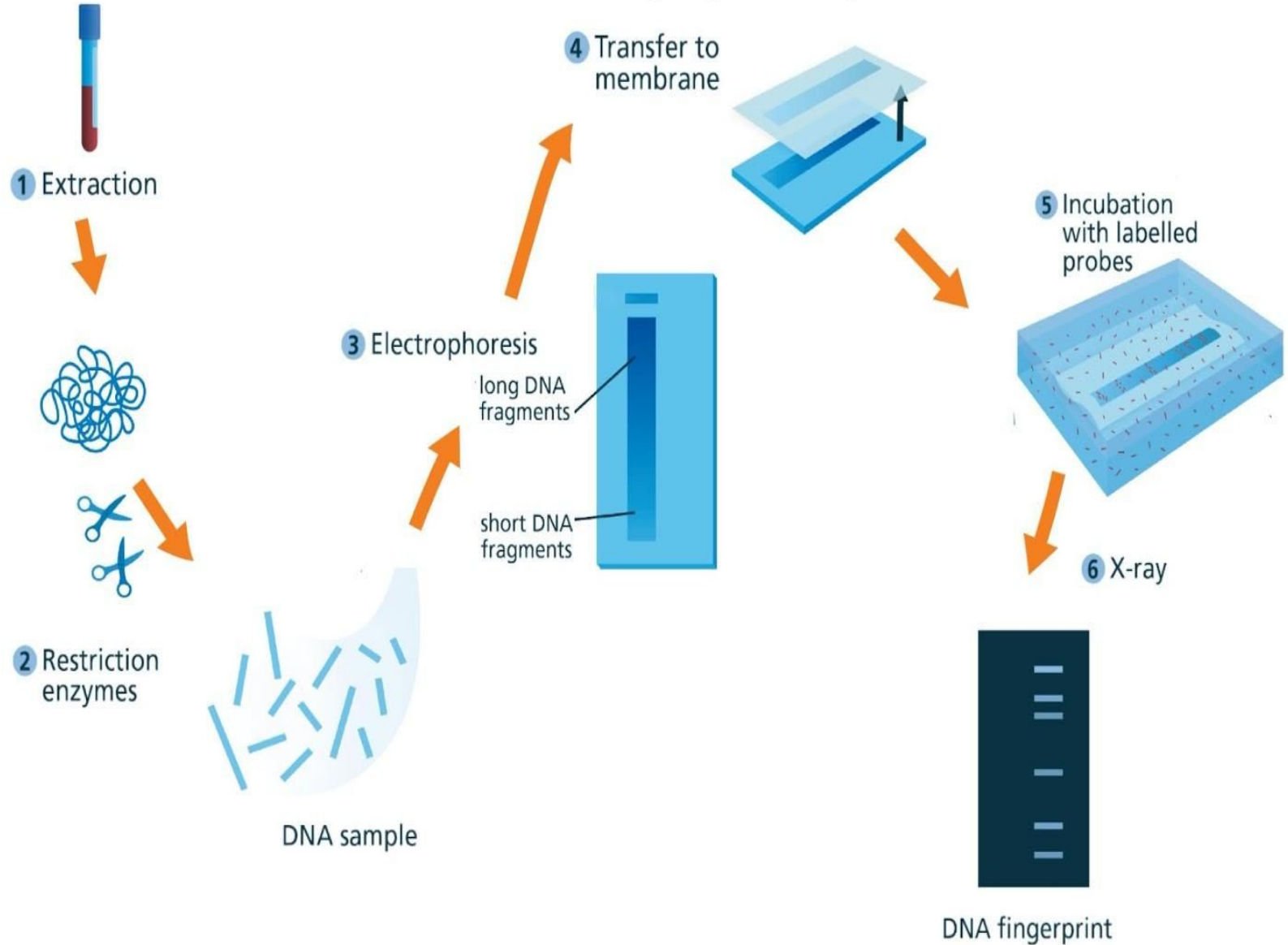
- **DNA Structure**

- The DNA structure can be thought of as a twisted ladder. This structure is described as a double-helix, as illustrated in the figure above. It is a nucleic acid, and all nucleic acids are made up of nucleotides. The DNA molecule is composed of units called nucleotides, and each nucleotide is composed of three different components such as sugar, phosphate groups and nitrogen bases.
- The basic building blocks of DNA are nucleotides, which are composed of a sugar group, a phosphate group, and a nitrogen base. The sugar and phosphate groups link the nucleotides together to form each strand of DNA. Adenine (A), Thymine (T), Guanine (G) and Cytosine (C) are four types of nitrogen bases.
- These 4 Nitrogenous bases pair together in the following way: **A** with **T**, and **C** with **G**. These base pairs are essential for the DNA's double helix structure, which resembles a twisted ladder.
- The order of the nitrogenous bases determines the genetic code or the DNA's instructions.
-

structure of DNA



Process Of DNA Fingerprinting



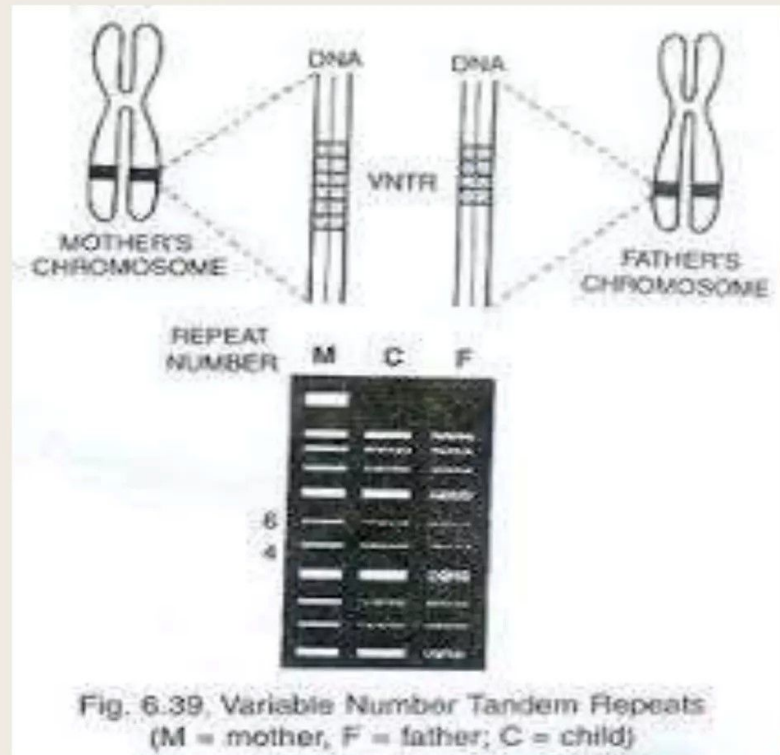
What is DNA Fingerprinting?

- DNA Fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA.
- The process of DNA fingerprinting was invented by Alec Jeffrey at the University of Leicester in 1985.
- Also called DNA Profiling or Molecular Fingerprinting.

PRINCIPLE

Variable Number of Tandem Repeats {VNTR}

- Small part of DNA vary from individual to individual
- Chances 30,000 million to 1 (except for identical twins).

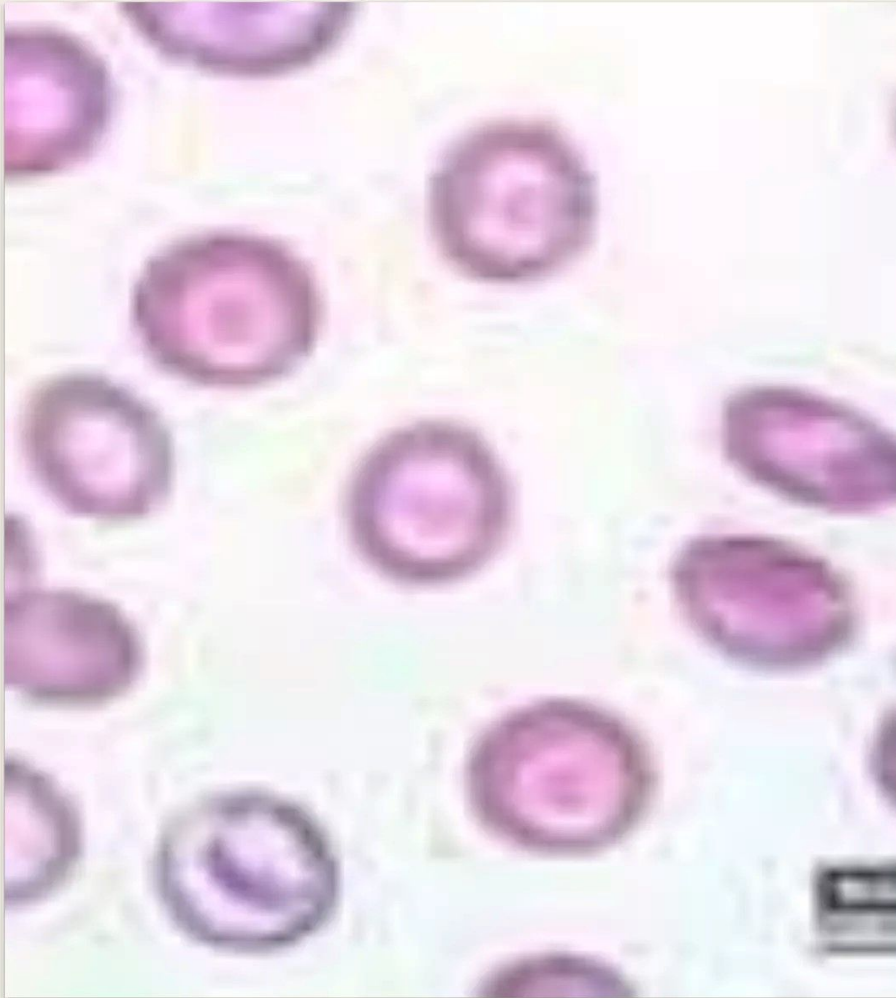


Methodology

Steps involved :-

1. DNA Extraction
2. DNA Cutting
3. Gel Electrophoresis
4. Southern Hybridization
5. Autoradiography

1. DNA EXTRACTION



- Cells are broken down to release DNA.
- Sample Collect from:-
 - ✓ Blood
 - ✓ Hair
 - ✓ Saliva
 - ✓ Semen
 - ✓ Body tissue cells

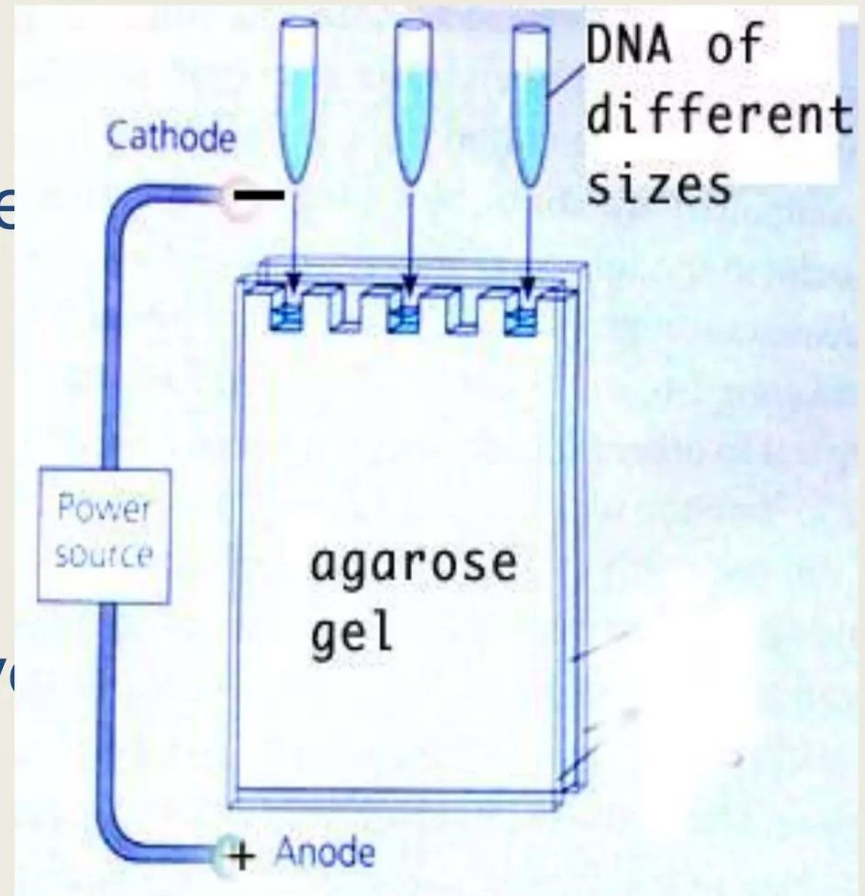
2. DNA CUTTING

- The DNA is cut into fragments using **restriction enzymes**.
- Each restriction enzyme cuts DNA at a specific base sequence.

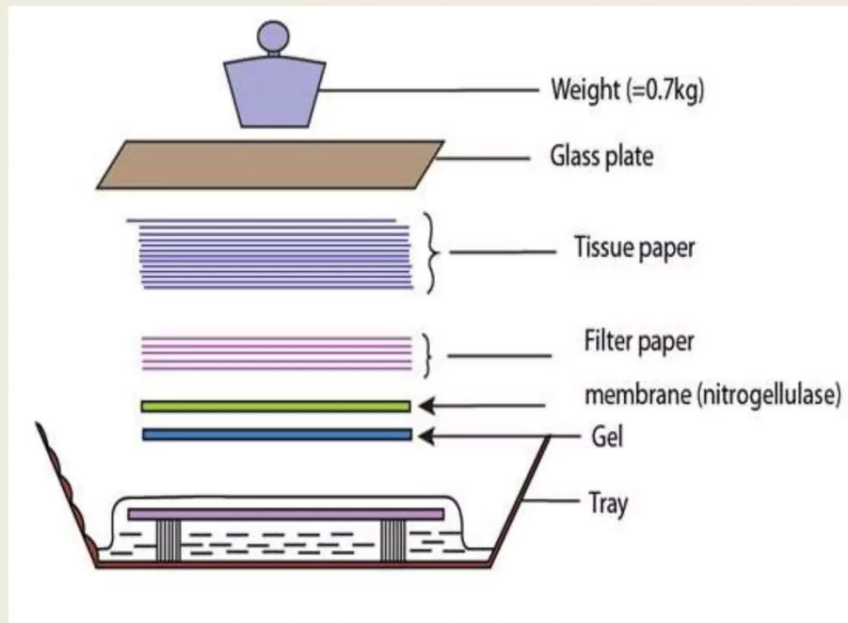


3. Gel Electrophoresis

- Fragments separated by length
- DNA (negatively charged)
- Moves towards +ve terminal
- Shorter fragments move faster

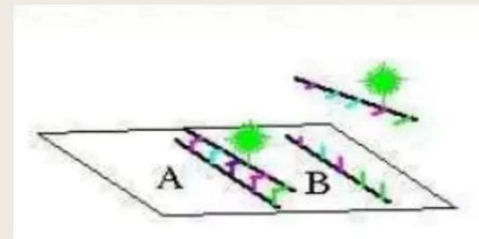


4. Southern Hybridization

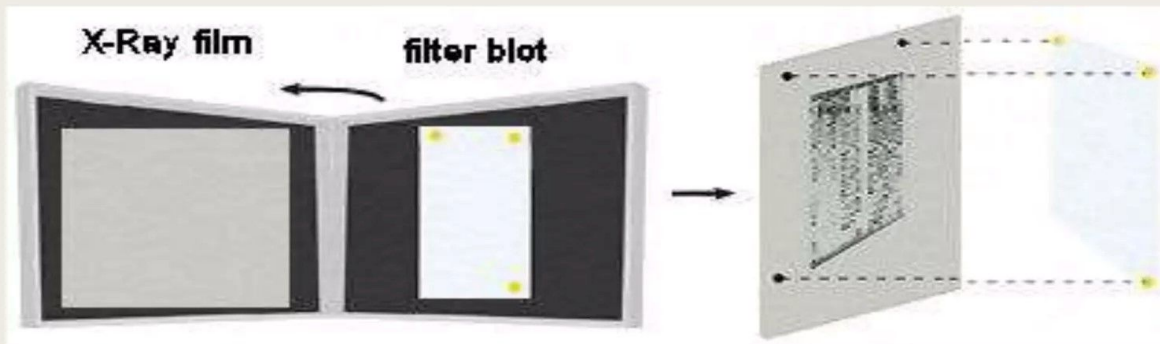


- DNA fragments transferred from gel to filter paper or nylon membrane
- DNA is split into single strands using an alkaline solution

- Radioactive probe in solution binds to DNA



5. Autoradiography



- X-ray film placed over filter paper.
- Radioactivity probes makes dark spots on film.

DNA Fingerprinting
patterns

